

MISSION 4

Part I. Examples

Example 1 : D_4 unfolding.

Example 2 : Hamiltonian system :

$$\dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

with $B_1 = 1, B_2 = 0$ or $B_1 = 0, B_2 = 1$

Part II. Checking examples

Example 1.

We have :

$$F(x, y) = \frac{1}{3}x^3 - xy^2 + \lambda_1(x^2 + y^2) + \lambda_2x + \lambda_3y$$

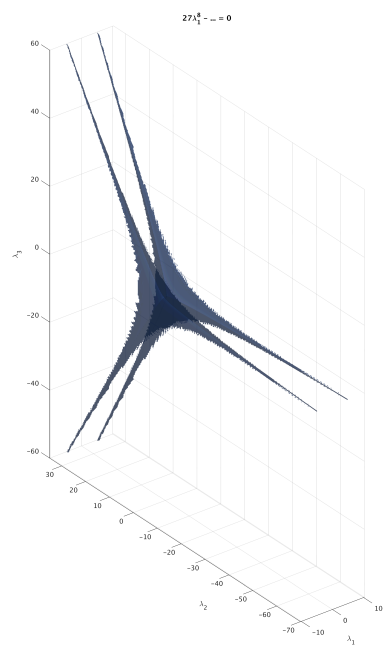
- Caustic(Σ_1) :

$$\exists (x_0, y_0) \in \mathbb{R}^2 \text{ s.t } \begin{cases} F_x(x_0, y_0) = 0 \\ F_y(x_0, y_0) = 0 \\ \text{Det}(\text{Hess}(H(x_0, y_0))) = 0 \end{cases}$$

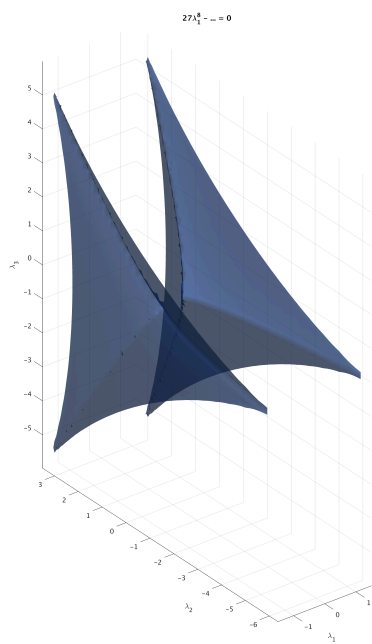
$\Rightarrow$

$$27\lambda_1^8 - 18\lambda_1^4\lambda_2^2 - 8\lambda_1^2\lambda_2^3 - \lambda_2^4 - 18\lambda_1^4\lambda_3^2 + 24\lambda_1^2\lambda_2\lambda_3^2 - 2\lambda_2^2\lambda_3^2 - \lambda_3^4 = 0 \quad (*)$$

The graph is like :



around (0,0,0) is like :



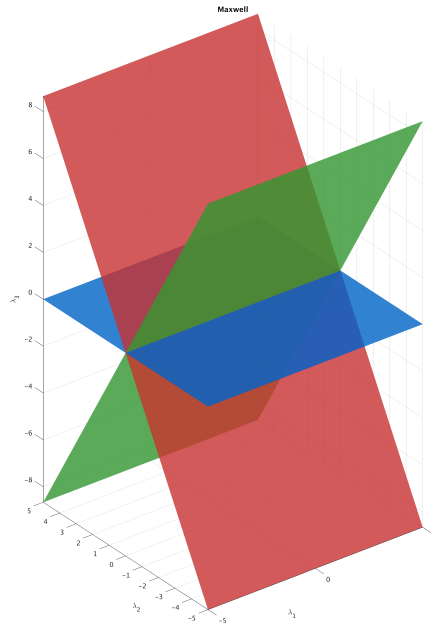
- Maxwell stratum(Σ_2) :

$\exists$ different $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ s.t
$$\begin{cases} F_x(x_1, y_1) = 0 \\ F_x(x_2, y_2) = 0 \\ F_y(x_1, y_1) = 0 \\ F_y(x_2, y_2) = 0 \\ F(x_1, y_1) = F(x_2, y_2) \end{cases}$$

$\Rightarrow$

$$\lambda_3(\lambda_3^2 - 3\lambda_2^2) = 0 \tag{**}$$

The graph is like :



Example 2.

- I. $B_1 = 1, B_2 = 0$

$$H(x, y) = -\frac{1}{2}(x^2 + y^2) + (1 - A_1)x^2y - \left(\frac{1}{3} + A_1\right)y^3$$

By the definition of Caustic(Σ_2)

$\Rightarrow$

$$-3 < A_1 < 1 \quad (***)$$

- II. $B_1 = 0, B_2 = 1$

$$H(x, y) = -\frac{1}{2}(x^2 + y^2) - A_1x^2y - A_1y^3 + xy^2 - \frac{1}{3}x^3$$

By the definition of Caustic(Σ_2)

$\Rightarrow$

$$A_1 = 0 \quad (****)$$

Part III. Code

- Code of (*) is :

```
[INPUT] :
clearAll["Global`*"];
```

```

Clear[\[Lambda]1, \[Lambda]2, \[Lambda]3, x, y]
F = 1/3 x^3 - x y^2 + \[Lambda]1 (x^2 + y^2) + \[Lambda]2 x + \[Lambda]3 y;
Fx = D[F, x];
Fy = D[F, y];
hess = {{D[Fx, x], D[Fx, y]}, {D[Fy, x], D[Fy, y]}};
deth = Det[hess] // Simplify;
eqns = {Fx == 0, Fy == 0, deth == 0};

GroebnerBasis[eqns, {\[Lambda]1, \[Lambda]2, \[Lambda]3}, {x, y}]

[OUTPUT] :

{27 \[Lambda]1^8 - 18 \[Lambda]1^4 \[Lambda]2^2 -
8 \[Lambda]1^2 \[Lambda]2^3 - \[Lambda]2^4 -
18 \[Lambda]1^4 \[Lambda]3^2 +
24 \[Lambda]1^2 \[Lambda]2 \[Lambda]3^2 -
2 \[Lambda]2^2 \[Lambda]3^2 - \[Lambda]3^4}

```

- Code of (**) is :

```

[INPUT] :

Clear[F, Fx, Fy];

F[x_, y_] := 1/3 x^3 - x y^2 + l1 (x^2 + y^2) + l2 x + l3 y;
Fx[x_, y_] := Evaluate[D[F[x, y], x]];
Fy[x_, y_] := Evaluate[D[F[x, y], y]];

(*t makes (x1,y1) , (x2,y2) different*)
polys = {Fx[x1, y1], Fy[x1, y1], Fx[x2, y2], Fy[x2, y2],
F[x1, y1] - F[x2, y2], t ((x1 - x2)^2 + (y1 - y2)^2) - 1};

(*candle out x1,y1,x2,y2,t , stay l1,l2,l3*)
relations = GroebnerBasis[polys, {l1, l2, l3}, {x1, y1, x2, y2, t}];

Factor /@ relations // Column

[OUTPUT] :

l3 (3 l2^2 - l3^2)

```

- Code of (***) is :

```
[INPUT] :

ClearAll["Global`*"];

F[x_, y_] = -1/2 (x^2 + y^2) + (1 - A1) x^2 y - (1/3 + A1) y^3;
Fx[x_, y_] = D[F[x, y], x];
Fy[x_, y_] = D[F[x, y], y];

cond = Fx[x1, y1] == 0 && Fy[x1, y1] == 0 && Fx[x2, y2] == 0 &&
Fy[x2, y2] == 0 && F[x1, y1] == F[x2, y2] && (x1 != x2 || y1 !=
y2);

Reduce[Exists[{x1, y1, x2, y2}, cond], A1, Reals]

[OUTPUT] :

-3 < A1 < 1
```

- Code of (****) is :

```
[INPUT] :

ClearAll["Global`*"];

F[x_, y_] := -1/2*(x^2 + y^2) - A1*x^2*y - A1*y^3 + x*y^2 -
1/3*x^3;
Fx = D[F[x, y], x];
Fy = D[F[x, y], y];

(*equations system*)
(*s1,s2 to make (x1,y1) , (x2,y2) different*)
eqs = {(Fx /. {x -> x1, y -> y1}) == 0, (Fy /. {x -> x1, y -> y1})
==
0, (Fx /. {x -> x2, y -> y2}) == 0, (Fy /. {x -> x2, y -> y2})
==
0, F[x1, y1] == F[x2, y2], s1*(x1 - x2) + s2*(y1 - y2) == 1};

(*GroebnerBasis*)
polys = {Fx /. {x -> x1, y -> y1}, Fy /. {x -> x1, y -> y1},
Fx /. {x -> x2, y -> y2}, Fy /. {x -> x2, y -> y2},
F[x1, y1] - F[x2, y2], s1*(x1 - x2) + s2*(y1 - y2) - 1};

A1Relation = GroebnerBasis[polys, {A1}, {x1, y1, x2, y2, s1, s2}];
A1Relation

[OUTPUT] :

{A1}
```